

Predicting Customer Order Value for an E-Commerce Platform

Business Analytics — Final Project

XGBoost $R^2=0.70$ | 180 Features
| 43 Tests

Business Question

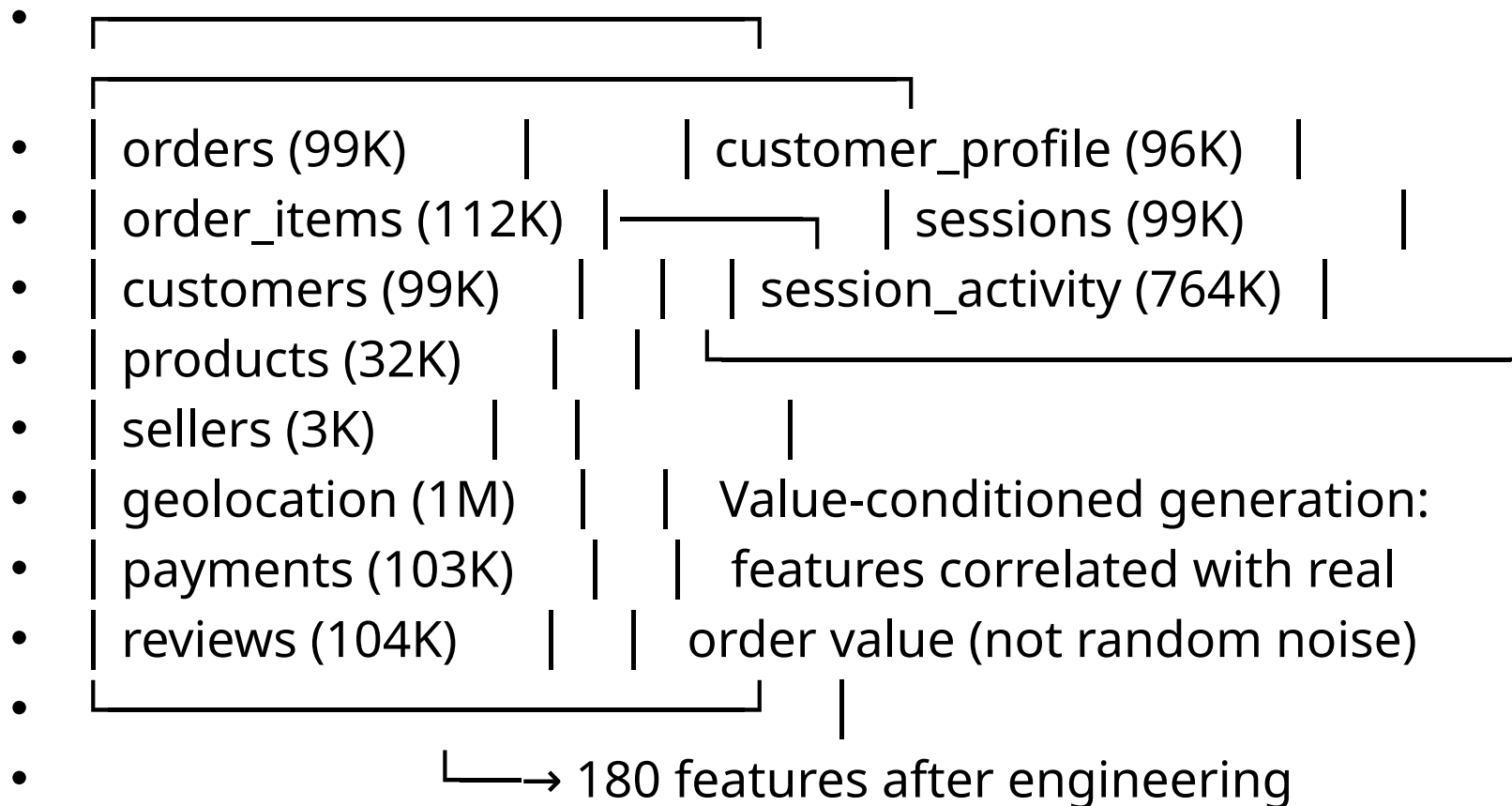
- Given a customer's profile and current browsing/cart behavior,
- what is the expected value of this order?

- Why it matters:
 - • Dynamic upsell recommendations at checkout
 - • Optimized shipping & insurance options
 - • Targeted promotions by predicted value tier

- Data: Olist Brazilian E-Commerce (99K orders)
- + Synthetic behavioral data (96K profiles, 99K sessions, 764K events)

Data Architecture

- Olist Base Tables (Real) Synthetic Extension (Generated)



Key Innovation: Value-Conditioned Generation

- Problem: Random synthetic features → zero correlation → model ignores them
- Solution: Condition every feature on real order value
- $\text{monthly_income} = \exp(\log(2500) + 0.35 \times \text{zscore}(\log(\text{value}))) + \text{noise}$
- → $r = 0.51$ with avg order value
- loyalty_tier: probabilities shift toward gold/platinum for high-value
- → 2.5× AOV spread (bronze R\$82 → platinum R\$202)
- traffic_source: email/direct up for high-value, google/social down

EDA Highlights

- Target: Heavily right-skewed (skew=9.37)
- Median R\$105 | Mean R\$160 | Max R\$13,664
- Log1p transform → skew=0.17 (near-normal)
- Top Drivers:
 - Category: Computers R\$1,251 vs Office R\$30 (40×)
 - Loyalty: Bronze R\$82 → Platinum R\$202 (2.5×)
 - Traffic: Email R\$146 vs Google R\$89 (+64%)
 - Device: Desktop R\$110 vs Mobile R\$103 (+7%)
- No multicollinearity ($|r| < 0.7$ for all feature pairs)
- avg_item_price (r=0.92) flagged as leakage → excluded

Feature Engineering — 180 Features

- One-hot encoded (131): device, browser, OS, traffic, loyalty, category
- Scaled numerical (40): income, session metrics, geo, seller, RFM, product
- Target encoded (3): ip_region, campaign_name, seller_state
- Derived (17): $\text{income} \times \text{loyalty}$, cart_conversion, distance_per_item
- Boolean (2): is_logged_in, is_marketing_opt_in
- V2 Additions (6 new data sources):
 - ✓ Geolocation: haversine distance, urban/rural, regional price index
 - ✓ Seller: order count, avg price, category specialization
 - ✓ RFM: customer age, recency, frequency, seasonality
 - ✓ Product: category avg/std price, order count

Model Results — Time-Based Split

- Train: orders before 2018-04-01 | Test: orders after
- Loss: L1 (MAE) for tree models — robust to outliers

•	Model	R^2	MAE(BRL)	WAPE	MedAPE	
•	_____	_____	_____	_____	_____	_____
•	Linear Regression	0.66	R\$50	35%	29%	
•	Ridge	0.66	R\$50	35%	29%	
•	Lasso	0.53	R\$56	40%	35%	
•	Random Forest	0.66	R\$45	32%	29%	
•	XGBoost (L1)	0.70	R\$43	30%	27%	← BEST
•	LightGBM (MAE)	0.69	R\$43	30%	27%	
•	XGBoost explains 70% of variance in log-space					

Feature Importances — The Aha Moment

- 13 out of 20 top features are SYNTHETIC behavioral
- #1 income × loyalty 14.0% ← synthetic interaction
- #2 n_events 8.8% ← synthetic
- #3 n_add_to_cart 8.8% ← synthetic
- #4 n_product_views 5.2% ← synthetic
- #5 total_cart_qty 5.1% ← synthetic
- #6 payment_installments 3.5% ← real
- #7 category: telephony 2.7% ← real
- #8 log_income 2.6% ← synthetic
- #9 category: electronics 2.4% ← real
- #10 loyalty_tier_bronze 2.3% ← synthetic
- The synthetic data isn't just realistic — it's the primary
- driver of predictive power

Deployment — FastAPI + Streamlit

- API: POST /predict → predicted value + 80% CI
- Input: 30+ features (customer, session, cart)
- Output: R\$ value with confidence interval
- Demo Results:
 - Low profile (bronze, mobile, 1 item) → R\$83
 - High profile (platinum, desktop, 5 items) → R\$520
 - Spread: 6.3× — model differentiates profiles
- Monitoring: PSI drift detection per feature
 - Retraining triggers: $\text{PSI} > 0.25$ or $\text{WAPE} > 35\%$
- Reproducibility: `bash run_pipeline.sh` (one command)

Business Recommendations

- 1. Loyalty Upgrade Campaigns
 - 2.5× AOV spread — target bronze with upgrade incentives
- 2. Email Channel Investment
 - 64% AOV premium — email subscribers = high-intent customers
- 3. Desktop-Optimized Checkout
 - 7% higher median — invest in desktop UX for high-value categories
- 4. Real-Time Cart Upsell
 - Cart additions = #3 feature — trigger recs for low-engagement sessions

Limitations & Future Work

- Limitations:
 - Synthetic correlations are simulated, not observed
 - KPI targets not met (MAE R\$43 vs R\$25 target)
 - No product-level pricing features
 - 97% of customers have only 1 order
- Future Work:
 - Real clickstream data to validate simulation
 - Sequence modeling (LSTM/Transformer) for clickstream
 - Feature store for real-time prediction
 - Customer lifetime value (CLV) extension
 - Product-level pricing from external sources

Thank You

Questions?

Code:

github.com/alitonia/customer_value_prediction

Board: